

ALLEGHENY MILLWORK

TORNADO PREPAREDNESS PLAN & CHECKLIST

Interior Finish Trades – Division 06: Rough & Finish
Carpentry/Architectural Millwork

Interior Trim | Casework | Paneling | Stairwork | AWI/AWS Architectural Woodwork

Project Name & Number:	
Project Address:	
General Contractor:	
Plan Prepared By:	
Date Issued:	
Revision/Version:	

TORNADO PREPAREDNESS PLAN & CHECKLIST

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PURPOSE & SCOPE

This Tornado Preparedness Plan establishes the procedures, responsibilities and protective measures that Allegheny Millwork requires to safeguard personnel, architectural millwork, interior carpentry materials and completed work from a tornado and severe thunderstorm events. This plan applies to all work performed under CSI MasterFormat Division 06 Sections 06 20 00 (Finish Carpentry) and 06 40 00 (Architectural Millwork).

This plan is intended to comply with OSHA 29 CFR 1926 (Construction Industry Standards), OSHA's General Duty Clause regarding recognized severe weather hazards, NFPA 1600 (Standard on Continuity, Emergency & Crisis Management), FEMA P-361 (Safe Rooms for Tornadoes & Hurricanes), FEMA P-431 (Tornado Protection – Selecting Refuge Areas in Buildings), the AWI Quality Standards for Architectural Woodwork, and the project's governing contract documents.

NOTE: Tornadoes differ fundamentally from hurricanes in warning times. Where a hurricane allows 72 – 96 hours of preparation, a tornado may only give 5 – 15 minutes between a Warning and Impact. This plan is therefore built around pre-staged protections, rapid response actions and, above all, personnel sheltering.

APPLICABLE DIVISION 06 SCOPE SECTIONS

- 06 20 00: Finish Carpentry (Interior Trim, Running & Standing Trim, Millwork Installation)
- 06 22 00: Millwork (Shop-Fabricated Interior Woodwork)
- 06 25 00: Prefinished Paneling
- 06 26 00: Board Paneling
- 06 40 00: Architectural Woodwork (AWI/AWS Premium, Custom & Economy Grade)
- 06 41 00: Architectural Wood Casework
- 06 42 00: Wood Paneling (flush, stile-and-rail, ornamental)
- 06 43 00: Wood Stairs and Railings
- 06 44 00: Ornamental Woodwork
- 06 46 00: Wood Trim
- 06 48 00: Wood Frames (architectural wood door frames)

WHY INTERIOR CARPENTRY & MILLWORK REQUIRE A DISTINCT PLAN

Interior carpentry and architectural millwork represent some of the highest dollar-per-square-foot material values on a project. A single-veneered conference room package or a run of AWI Premium Grade paneling can exceed the cost of the structural frame it attaches to. These materials share specific vulnerabilities that a generic site plan does not address:

- **Moisture Intolerance:** Water contact for even a few hours can cause veneer delamination, panel warping, and adhesive failure. Damaged millwork is generally non-repairable under AWI standards

- Environmental Sensitivity: AWI requires 25–55% relative humidity and 60–90°F; a broken envelope during a storm can swing conditions outside this range in minutes
- Schedule Exposure: Millwork is typically installed late in the project when the building is weathertight; a tornado event after millwork delivery but before substantial completion creates maximum financial exposure
- Debris Impact: Wind-borne debris penetrating a single window can damage an entire room's worth of installed paneling
- Irreplaceability: Many veneer flitches and custom profiles cannot be re-sourced; a damaged item may force a complete re-design

PLAN ACTIVATION & ALERT LEVELS

Activation is tied to products issued by the National Weather Service (NWS) Storm Prediction Center (SPC) and the local NWS Weather Forecast Office. All Allegheny Millwork Superintendents working on a jobsite that has the potential for severe weather impacts, shall maintain at least two independent means of receiving alerts:

1. NOAA Weather Radio with SAME programming for the project county
2. A Cellular alert app such as FEMA, WEA or an equivalent

Alert Level	Trigger	Timeframe	Primary Division 06 Action
Level 1: Seasonal Awareness	- Convective Season Active - Primary: March – June - Secondary: November - SPC Outlook Day Issued	Days to Weeks	- Maintain Pree-Staged Protection Kits - Verify NOA Radios - Confirm Shelter Designations
Level 2: Severe T-Storm Watch or Tornado Watch	SPC Issues Watch Covering Project County	2 – 8 Hours	- Pre-Position Tarps - Move Loose Millwork Deliveries to Interior Staging - Brief Crew on Shelter Locations
Level 3: Severe T-Storm Warning	NWS WFO Issues Warner for Project Area	30 – 60 Minutes	- Cover High-Value Millwork with Pre-Staged Poly - Suspend Exterior Work - Move to Interior of Building
Level 4: Tornado Warning	- NWS WFO Issues Warning - Confirmed Funnel/Rotation Reported	0 – 15 Minutes	- IMMEDIATE SHELTER - All Personnel to Designated Refuge Area - No Protective Actions Delay Sheltering
Level 5: All Clear	- NWS Cancels Warning - Prime Safety Officer Authorized Reentry	Post-Event	- Personnel Accountability - Damage Assessment - Material Disposition - Moisture Testing

CRITICAL RULE: At Level 4 (Tornado Warning), personnel sheltering is the ONLY priority. Do not attempt to cover, relocation or protect any material once a Warning is issued for the project area. Material is Replaceable. Employees are NOT.

DISTINGUISHING WATCHES FROM WARNINGS

- **Watch:** Conditions are favorable for development. This is a preparatory signal, not an immediate threat signal.
- **Warning:** A severe thunderstorm or tornado has been detected by radar or observed by spotters. Take protective action immediately.
- **Tornado Emergency:** A variant Warning used when a large, damaging tornado is confirmed and populated areas are in its path. Treat as Level 4 with maximum urgency.

SHELTER DESIGNATION & SHELTERING PROTOCOL

The single most important element of tornado preparedness on a construction site is identifying a refuge area before the season begins and briefing every worker on its location. FEMA P-4321 provides the guidance that Allegheny Millwork uses here.

SHELTER HIERARCHY (BEST TO WORST)

1. ICC 500/FEMA P-361 complaint storm shelter, if present on or adjacent to the project
2. Below-Grade space in a completed or substantially completed concrete or masonry structure, typically interior rooms away from windows.
3. Interior room on the lowest occupied floor of a substantially completed structure. Small rooms with short wall spans (restrooms, interior corridors) are preferred. Avoid rooms with long-span roofs.
4. Adjacent completed structure (coordinate pre-event with that building's management).
5. Reinforced onsite storm refuge (rare on most projects).

LOCATIONS NEVER TO USE AS SHELTER

1. Temporary Site Trailers & Modular Offices: These offer essentially zero protection and become projectiles.
2. Vehicles: Cars & pickup trucks are not safe shelter from tornadoes.
3. Storage Containers (Conex Boxes): Steel boxes roll and tumble in EF2+ winds.
4. Under Overpasses or Bridge Structures: These can funnel wind and are not safer than open ground.
5. Rooms with Exterior Windows or Long-Span Ceilings: Unframed lobby spaces, gymnasiums and atriums
6. Top Floor of Any Structure or rooms directly beneath long-span trusses.
7. Areas with stored sheet goods, panel stacks or heavy materials that could shift.

SHELTERING PROTOCOL

- At Level 4, move to a designated shelter by the shortest path. Do not stop to gather tools, phones or materials.
- Once sheltered, crouch low, protect head and neck with arms and if available, use a hard hat and a cushioning item, such as a jacket or drop cloth.
- Remain in shelter until the NWS cancels the Warning **and** the Prime Safety officer authorizes reentry – NOT when the noise stops (multiple circulations are in motion).
- Do not attempt to rescue any damaged areas until the Safety Officer and emergency services have cleared the structure.

SHELTER DESIGNATION RECORD

The following shall be recorded in Appendix A and posted at every crew must point:

Primary Shelter Location	
Capacity (Persons)	
Secondary Shelter Location	
Shortest Route from Work Area	
Maximum Travel Time (Minutes)	
Last Walkthrough/Verification Date	

ROLES & RESPONSIBILITIES

PROJECT MANAGER

- Overall responsibility for plan execution
- Maintains NOAA Weather Radio with SAME coding for the project county
- Calls alert-level escalation and authorizes work stoppage
- Ensures every employee, subcontractor and installer has been briefed on shelter location and route within 24 hours of mobilization.

MILLWORK INSTALLATION LEAD

- Maintains the inventory of on-site architectural millwork with location and value
- Pre-Stages protection kits (poly, type, padding) at each active installation area
- Executes Level 3 coverage actions on installed and staged millwork
- Documents pre-event condition with photographs keyed to shop drawing references

SITE SAFETY COORDINATOR

- Monitors active weather products and escalates alert level to PM
- Conducts shelter drill mobilization and again at the start of each severe weather season
- Confirms every worker's cellular device is configured for Wireless Emergency Alerts (WEA)
- Performs head count at shelter, reports missing workers to incident command

ENVIRONMENTAL CONTROL LEAD

- Monitors interior relative humidity and temperature in areas containing installed millwork
- Coordinates with MEP and General Conditions trades for temporary conditioning
- Executes rapid envelope dry-in if the building is breeched during the event

MATERIAL SPECIFIC PROTECTION MEASURES

Protection strategy depends on the product's state (stored, staged for install, or installed) and its sensitivity to moisture and impact. Given tornado warning times, all protections below must be capable of deployment in under 30 minutes with pre-staged materials.

ARCHITECTURAL WOODWORK IN INTERIOR STORAGE

AWI/AWS compliance requires stable environmental conditions. Tornado events threaten this in two ways: Direct impact damage and envelope compromise that allows humidity/temperature excursions.

- Store all millwork in the interior core of the building, away from exterior walls and glazing
- Elevate on pallets or dunnage a minimum of six (6) inches above slab to avoid ground water migration
- Cover each staged item with a breathable woven tarp (Thermo-Lite or equivalent); avoid direct poly contact which traps condensation and can ghost veneer finishes
- Over the breathable cover, drape a secondary 6-mil poly sheet that can be quickly deployed at Level 3 to shed water
- Keep each item's shop ticket, AWI QCP label and protection photograph bundled in a weatherproof sleeve attached to the item
- Never stage architectural millwork beneath a skylight, clerestory or in a room with long span roof trusses

WOOD CASEWORK (IINSTALLED & STAGED)

- Installed Casework with Exposed Interiors: pre-stage pre-cut cardboard corner guards and padded furniture blankets at each run
- Installed Casework: at Level 3, close all doors and drawers; drape the run with moving blankets then a poly sheet; weight the poly at the floor with sandbags or unopened material cartons
- Staged Casework Awaiting Installation: keep crated or on pallets; do not uncrate until ready to set
- Countertops (Solid Surface, Stone Composite, Wood): store flat, fully supported; a damaged or warped top cannot be straightened
- Critical Detail: do not attach protective coverings to finished surfaces with adhesive tape. Use elastic banding, strapping, or weighted drop cloths

WOOD PANELING (FLUSH, STILE-AND-RAIL, ORNAMENTAL)

Installed paneling presents the single highest field risk in Division 06 interior work. Once mounted, it cannot be relocated and protective coverage is the only option.

- Pre-Fabricate & Pre-Size Paneling Coverage Kits at Project Mobilization: roll out a full-height sheet of 6-mil poly for each paneled wall, pre-labeled by room and wall

- At Level 3, Deploy Coverage Kit: hang poly from temporary wood furring or adhesive hooks installed at ceiling grid, allow poly to drape across face of paneling, weight bottom with sandbags at floor
- Never Use Painter's Tape or Masking Tape Directly on Finished Panel Surfaces: tape residue damages factory finish
- For Matched-Grain Veneer Paneling: photograph each wall pre-event at high resolution with labeled key elevations; a single damaged panel may require re-sequencing from the original flitch, which may no longer be available

WOOD STAIRS, RAILINGS & STANDING/RUNNING TRIM

- Installed Stair Parts (Treads, Risers, Stringers, Rails, Balusters): cover with ram board, then drape with moving blankets, then poly
- Protect handrail terminations with fitted caps or padded sleeves
- Running Trim (Base, Casing, Crown) at Vulnerable Locations Such as Under Windows: drape with poly from top of trim to floor line
- Standing Trim (Paneled Jambs, Pilasters, Column Wraps): wrap with padded blankets and secure with elastic banding, never tape
- Staged Trim Packages: keep in original banding, stacked on edge if possible, on dunnage in the building core

ARCHITECTURAL WOOD DOORS & FRAMES

- Pre-Hung Doors: keep in factory packaging until ready to set; factory packaging is optimized for protection and should not be prematurely discarded
- Installed Doors: close and latch at Level 3; tape painter's plastic or a cardboard face protector to the frame only, never to the door face
- Veneer doors are particularly sensitive to humidity swings and will warp if one face is exposed to elevated humidity while the other remains conditioned, maintain equal conditions on both faces
- Wood Frames (Pre-Finished or Field-Finished): protect jambs with corner guards; if doors are not yet hung, install temporary plywood closure panels in the opening at Level 3

ENVIRONMENTAL CONTROL DURING & AFTER THE EVENT

AWI Section 2 requires that architectural woodwork be maintained between 25–55% relative humidity and 60–90°F. A breached envelope during a tornado can violate both limits within minutes.

- Before the season, pre-position battery-backup hygrothermographs or data loggers in each room containing installed millwork; record baseline readings
- Pre-stage roll goods of 6-mil poly, furring strips, and cap nails at each stairwell for rapid envelope dry-in

- Confirm availability of portable dehumidifiers and air movers through the Prime Contractor or a pre-selected restoration vendor
- After any envelope breach: dry-in within 4 hours; restore conditioning within 24 hours; record humidity and temperature at 4-hour intervals until stable
- If humidity exceeds 60% for more than 48 hours in a room containing installed veneer millwork, coordinate with the architect and woodwork supplier regarding potential rejection

PRE-STAGED PROTECTION KITS

Because tornado warning times are measured in minutes, not hours, protection materials must be cached at the point of use before the season begins. A worker should not have to travel more than 100 feet from any active work or storage area to reach a protection kit.

ROOM-LEVEL KIT (ONE PER ACTIVE MILLWORK ROOM)

- Two pre-cut sheets of 6-mil poly, sized to the largest wall in the room
- Four moving blankets, folded and bagged
- Six sandbags (25 lbs. each) in a labeled tote
- One roll of ram board, pre-cut to floor dimensions
- Elastic banding, minimum of 50 feet
- Labeled room photograph card showing where protections are to be deployed
- Laminated quick-reference card with plan extract and shelter route

FLOOR-LEVEL KIT (ONE PER OCCUPIED FLOOR)

- Rolls of 6-mil poly (20' x 100'), minimum two rolls
- Furring strips and cap nails for emergency dry-in
- Hammer, utility knife, and staple gun in a labeled caddy
- Battery-operated work lights (minimum four)
- NOAA Weather Radio with SAME, fully charged or on mains power
- First-aid kit meeting ANSI Z308.1 Class B

INSPECTION & REPLENISHMENT

- Inspect all kits at the start of each calendar month during active season (March through June and November)
- Replace any consumed, damaged or degraded items within 48 hours
- Rotate battery powered items at least quarterly, date stickers required

PERSONNEL SAFETY, TRAINING & DRILLS

- New-hire orientation for every Division 06 worker shall include plan review, shelter walk-through, and signed acknowledgment within the first shift
- Shelter drills shall be conducted at least twice per severe-weather season; document date, time, and participating crew
- Target shelter travel time is 90 seconds from the furthest occupied work area; if exceeded, the shelter designation shall be reviewed
- Workers with mobility, hearing, or other necessary accommodations shall have an individual assisted-shelter plan identified in Appendix A
- Language-appropriate plan summaries shall be posted for any crew with limited-English-proficiency workers; do not rely solely on verbal briefings
- Severe-weather topics shall be added to the weekly tailgate agenda during peak season

DOCUMENTATION & INSURANCE REQUIREMENTS

Architectural millwork carries unusually high dollar value and long replacement lead times; insurance recovery depends on documentation of pre-event condition.

- High-resolution photographs of every installed millwork run, keyed to room number and shop drawing reference
- Moisture meter readings (pin-type and capacitance) for all installed veneer work, recorded monthly during storage and weekly once installed
- Environmental data logger records (temperature and humidity) for each room with installed architectural woodwork
- Copies of shop drawings, AWI QCP certificates, and supplier shipping tickets retained in the project record
- Builder's risk policy number, carrier, and adjuster contact posted in the site office and the Division 06 trailer
- Signed and dated copies of the Section 10 checklists retained in the project record
- Mobilization photograph set repeated at substantial completion of each millwork phase

POST-EVENT ASSESSMENT & RECOVERY

No personnel shall reenter any damaged area until structural, electrical and gas hazards have been cleared by the Prime Security officer and Local Authorities.

INITIAL RESPONSE (FIRST 2 HOURS)

1. Personnel accountability headcount at shelter, report any missing worker(s) to the Prime Incident Commander
2. Administer first aid and request EMS as needed; this precedes any damage assessment
3. Remain in or near the shelter until the Prime Security Office authorizes movement; secondary circulations are common
4. Avoid moving through damaged areas except along routes cleared by the Safety Officer

INITIAL DAMAGE ASSESSMENT (FIRST 24 HOURS)

1. Walk the Division 06 scope with the Safety Officer, identify envelope breaches first (water intrusion clock starts at breach, not at entry)
2. Photograph all damage in place before moving any material
3. Identify water-exposed and debris-impacted millwork, segregate from undamaged stock
4. Deploy emergency envelope dry-in at any breach in a room containing installed millwork
5. Notify builder's risk insurer within the policy reporting window (typically 48-72 hours)
6. Notify architectural woodwork supplier immediately; long lead times may need reordering before insurance adjudication completes

MATERIAL DISPOSITION

1. Veneer paneling and casework with any water intrusion: presumptively rejected; AWI QCP does not recognize field-dried water-damaged veneered products
2. Solid-wood trim and standing trim with water exposure: moisture-test; may be salvageable if dried within acceptable timeframe and below manufacturer limits
3. Plastic laminate casework: inspect for edge delamination; reject if any delamination is present
4. Architectural wood doors: check for warp, telegraphing of core defects, and finish damage; any veneer separation or measurable warp triggers rejection
5. Debris-impacted items: even minor impact can cause hairline cracks in veneer that will propagate over time; document and consult architect and supplier
6. Do not dispose of any damaged material before the insurer's adjuster has documented or waived inspection

DRYING & ENVIRONMENTAL RESTORATION

1. Follow IICRC S500 (Water Damage Restoration) for all wetted areas containing Division 06 work
2. Target: restore interior conditions to 25–55% RH and 60–90°F within 72 hours
3. Deploy commercial dehumidifiers and air movers; target grains per pound (GPP) below 75 for restoration drying

4. Document humidity and temperature at 4-hour intervals at multiple locations in each affected room until stable for at least 72 hours
5. Do not install replacement millwork until environmental conditions have been stable and documented for a minimum of 14 days

CHECKLISTS

The following checklists shall be completed, signed and retained in the project record. The Pre-Season, Watch and Warning Checklists are each time-boxed; they must be executable within the stated timeframe without exception.